PUNE INSTITUTE OF COMPUTER TECHNOLOGY

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PYTHON BOOTCAMP CAPSTONE PROJECT REPORT

Smart Cities: Identifying the "Last Mile" Bottleneck in Public Transport — A Commuter Behavioural-Switch Approach

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Abstract

Most people who use the metro or a city bus in India will admit, if asked honestly, that the part of the trip they dread is not the ride itself but the bit before and after it — getting from home to the station, and from the station to wherever they are actually headed. That stretch, usually a couple of kilometres at most, is what planners call the "last mile," and it is often the real reason someone decides to just take their own bike or car instead of bothering with public transport at all. A lot of the existing research on this problem looks at it from a fairly high altitude, using traffic volume or road density as the main signal. We wanted to try something a little closer to the ground: instead of asking how congested a corridor is on paper, we asked at which specific stations or stops commuters are actually giving up on transit, and what seems to be pushing them toward that decision. To explore this, we built a synthetic dataset styled after what a city transit authority or a mobility API such as Google Maps would realistically produce — trip-level details like wait time, crowding, how many auto-rickshaws or e-rickshaws are around, the weather that day, and whether the commuter has had a bad experience before. The eventual aim, still very much a work in progress, is to use this data to flag the handful of nodes where commuters are quietly defecting to private vehicles, and later to feed that information into a route-optimisation exercise for feeder buses.

Keywords— *last-mile connectivity, public transit, commuter behaviour, mode choice, congestion nodes, smart cities, synthetic dataset*

I. Introduction

Over the last decade or so, Indian cities have poured a lot of money and effort into mass rapid transit — metro lines, BRT corridors, suburban rail expansions, and so on. And to be fair, it has worked, at least on the trunk routes: once you are on the metro, travel time is usually predictable and fast. The trouble starts at the edges. Getting to the station in the first place, or figuring out how to cover the last stretch after you get off, is often clunky enough that people just give up and drive. This is not a new observation — it has been written about for years — but it is worth restating because it is the entire premise of this project.

What we are attempting here, as part of the Python Bootcamp capstone, is to look at this problem less like a road-engineering question and more like a behavioural one. Instead of asking "how bad is traffic near this station," we are asking "at which stations do people actually stop trusting public transport for their last mile, and why." That means paying attention to things like how long someone waits, how crowded the platform feels, whether it is raining, and whether there is a reasonably good auto or e-rickshaw around when they need one. It is a modest reframing, but we think it matters — and honestly, the modelling side of this is still very much under construction, so this report should be read as an early, careful first pass rather than a finished product.

II. Literature Review

We looked at two pieces of prior work while shaping this project, and they turned out to sit on opposite ends of the same problem — one zoomed in on the modes themselves, the other zoomed out to the whole network.

The first is a case study of last-mile connectivity around a mass rapid transit station in Gurugram [1]. It looks closely at how commuters actually get from the metro gate to their final stop — auto-rickshaws, e-rickshaws, feeder buses — and compares these options on things like how long you wait, what it costs, and how reliable they are. It is a genuinely useful reference for understanding what last-mile options typically exist on the ground and how people rate them. What it does not really get into is why one station sees far more people bailing out to private vehicles than another station with a broadly similar set of options — that comparison across stations isn't really the point of that paper.

The second is a working paper by Srinivasa [2] that tackles the problem from much higher up — it breaks a city's transit catchment area into spatial zones and studies how each zone's access characteristics feed into overall ridership and connectivity. This kind of framing is great if you are trying to plan interventions city-wide, but the unit being studied is still the zone or the corridor, not an individual commuter's decision at a particular stop on a particular evening.

Between the two, we felt there was a gap worth poking at: neither paper really sits at the level of a single trip, at a single node, asking what tips a person from "I'll wait for the bus" to "forget it, I'm taking my bike." That is the space this project is trying to occupy.

III. Problem Statement

Put simply: given data about individual trips and the stations they pass through, can we point to the specific high-congestion nodes where commuters are most likely to give up on public transport for the last stretch of their journey — and can we say something sensible about what is driving that choice at each of those nodes?

IV. Research Gap

A lot of the existing modelling around last-mile connectivity, the two papers cited above included, tends to be built around traffic volume, how much infrastructure exists, or how the network is laid out spatially. Those are all reasonable things to study, but they mostly answer “where is the network structurally weak,” not “why did this particular commuter, on this particular evening, decide public transport wasn't worth the hassle.” That second question is where we are trying to spend our effort — shifting the unit of analysis down from the corridor or the zone to the individual trip, and using things like wait time, crowding, weather, how many alternate modes were around, and whether the person had a bad experience before, as the ingredients for understanding that switch. We are hopeful, though we want to be honest that this is still an early hope rather than a proven result, that this kind of behavioural lens will surface interventions that a purely volume-based model would simply never notice.

V. Proposed Methodology

A. Data Source and Synthetic Dataset Generation

Ideally, this kind of analysis would run on real data — a City Transit Authority API for schedules, headways, and ridership counts, plus something like Google Maps mobility data for congestion and travel-time signals. We did not have access to either within the timeframe of the bootcamp, so instead of leaving that gap empty, we built a synthetic dataset in Python using NumPy and pandas that tries to mimic what those two sources would plausibly look like together. Rather than just generating random numbers in each column, we gave each of 120 imaginary transit nodes across six major Indian cities a few hidden, underlying traits — how congestion-prone it tends to be, how good the local autos/e-rickshaws/feeder buses are, and how decent the walking infrastructure is — and then simulated 10,000 individual trips on top of that, letting things like wait time, crowding, and weather vary in a way that is tied back to those node traits. The idea was to end up with something that actually has learnable structure in it, not just noise dressed up as a spreadsheet.

B. Feature Description

Broadly, the dataset we generated (last_mile_connectivity_synthetic.csv) is organised into the following groups of columns:

- Node identity and geography — city, transit_node_id, node_type, latitude/longitude.
- Trip context — trip_date, time_of_day, day_type, weather_condition, trip_purpose.
- Service-quality indicators — avg_wait_time_min, transit_headway_min, crowding_index, congestion_index_node.
- Last-mile options — alt_mode_availability_score, app_cab_wait_time_min, last_mile_fare_inr, walk_time_min, last_mile_infra_score.
- Commuter attributes — age_group, gender, income_bracket, previous_negative_experience, safety_perception_score.
- Outcome variables — mode_chosen_last_mile (categorical) and switched_to_private_vehicle (binary target).

C. Analytical Approach

Our plan, and we say plan deliberately because the model-building side of this is still underway and we would rather move through it carefully than rush a shaky result, is to start with straightforward exploratory analysis — profiling congestion and switch rates node by node (eda_analysis.ipynb) — before attempting anything more ambitious. From there, we want to build a ranking of nodes by their observed private-vehicle switch rate, so we have a shortlist of the worst offenders to focus on. Once that groundwork is solid, the next step is a classification model, most likely starting with logistic regression and moving to something like gradient-boosted trees if the data supports it, trained to estimate the probability that a given commuter at a given node and time will switch to a private vehicle. We are especially interested in what the feature importances from that model end up saying, since that is what should eventually feed into the route-optimisation work described in Section VIII. None of this is finished yet — we would rather report honest progress here than pretend the modelling is further along than it is.

VI. Dataset Description

Table I lists the columns we consider most important for the analysis, rather than the entire schema — the full dataset runs to 29 columns and 10,000 rows in total, and the complete list along with data types is kept in the accompanying README.md so this section doesn't turn into an exhaustive dump of column names.

TABLE I

KEY ATTRIBUTES OF THE SYNTHETIC LAST-MILE DATASET

Attribute	Type	Description
switched_to_private_vehicle	Binary (0/1)	Target label: whether the commuter used a private vehicle for the last-mile leg
congestion_index_node	Float [0–1]	Composite congestion level at the node for the given trip
crowding_index	Float [0–1]	Perceived/measured crowding at the node
avg_wait_time_min	Float (minutes)	Average wait time for onward transit/last-mile mode
alt_mode_availability_score	Float [0–10]	Availability/quality of autos, e-rickshaws, feeder buses at the node

VII. Expected Results and Preliminary Observations

Even at this early stage, running a first pass over the synthetic dataset gives us something to talk about. Roughly 28% of trips in the dataset end with the commuter switching to a private vehicle, and that switch is not spread evenly — it clusters heavily around nodes with high congestion and crowding scores, especially during the morning and evening peaks and when the weather is bad. Interestingly, nodes where alternate last-mile options (autos, e-rickshaws, feeder buses) are scarce show a much higher switch rate even when the average wait time itself isn't all that long, which hints that just having options nearby matters almost as much as how quickly they arrive. We want to be careful not to over-claim here since the underlying model is still being built, but these early patterns line up reasonably well with what the Gurugram case study [1] describes about mode availability, and they fit comfortably within the kind of node-level shortlist that the spatial-planning approach in [2] would find useful downstream.

VIII. Future Scope

Once the node ranking and the behavioural model behind it are on firmer footing — and we are treating that as a patient, iterative process rather than something to rush — the next real ambition for this project is an optimisation layer that can actually suggest how to adjust feeder-bus routes and frequencies at the nodes where people are defecting the most. We are thinking along the lines of a mixed-integer formulation, or possibly a heuristic approach such as a genetic algorithm or simulated annealing, that reallocates feeder-bus hours toward the nodes where it would do the most good per additional bus-hour spent. Longer term, we would also like to swap out the synthetic dataset for the real thing — actual City Transit Authority API data and Google Maps mobility data — once we can get access to it, and to broaden the behavioural model with seasonal patterns and socio-economic context that the current version doesn't yet capture. Incredibly, so with worthwhile API keys as well as json integration there is also a scope of integrating the LLM (large language models) in our models, to make it more interactive with the user so to efficiently and provokingly provide all the necessary details which may be required to user at a day or time instance. Making the model be very much precise and focuses on necessary analysis on given or used travelling route.

IX. Conclusion

What we set out to do here was reframe the last-mile problem as a question about commuter behaviour rather than just traffic volume — to ask not only where the network is weak, but where and why people are actually walking away from it. We built a synthetic dataset of 10,000 trips across 120 nodes to give ourselves something real to work with, and even the early patterns in it point toward wait time, crowding, and the availability of alternate modes as the factors that matter most. The model that will eventually sit on top of this data is still being built, deliberately and carefully rather than in a hurry, but we think the groundwork laid out in this report gives the project a solid and honest foundation to build the route-optimisation phase on.

References and Citations

[1] "Performance of Public Transit Modes for Last Mile Connectivity near the Mass Rapid Transit System — A Case Study of Gurugram City."

[2] M. Srinivasa, "A Spatial Decomposition Based Framework for Building Last Mile Connectivity to Urban Public Transit Systems," Working Paper 723.